Haozhe Lei

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EDUCATION

New York University Sept. 2022 – Expected May 2027
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS
Award: 2023 Ernst Weber Fellowship

New York University Sept. 2020 – May 2022
M.S. in Computer Engineering

China Agricultural University Sept. 2015 – Jun. 2019
B.E. in Electrical Engineering and Automation

RESEARCH INTERESTS

- **Methods:** spatially aware RF sensing and wireless localization, ISAC signal processing and inference, probabilistic uncertainty quantification, wireless digital twins and ray tracing, object-centric semantic 3D memory, foundation-model agents, physics-informed and neuro-symbolic RL
- **Systems and Applications:** uncertainty-aware 6G positioning and environment mapping, indoor robot navigation and search, wearable/AR spatial assistance, FR3/mmWave channel-sounding testbeds, multi-band UE antenna/beam management, closed-loop wireless control

RESEARCH EXPERIENCE

NYU WIRELESS *Brooklyn, NY, USA*
Research Assistant *Sept. 2022 – Present*

Research agenda: connecting belief, world models, and action—turning sparse RF, visual, and device observations into calibrated spatial beliefs through wireless digital twins, then using them for 6G and robotic decisions.

- **RF sensing and posterior localization**: Modeling wireless localization as a calibrated spatial posterior over candidate transmitter locations, retaining multimodal hypotheses caused by AoA/SNR ambiguity, receiver directivity, blockage, and reflected paths. Developed *MC-CLE* (Monte Carlo Candidate-Likelihood Estimation), which learns candidate likelihoods from single-dominant-path AoA/SNR and accumulates evidence across observations; extended it to *LOCUS-DT* (Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins), which scores measured/synthetic multipath compatibility against a ray-tracing candidate library for site-agnostic, layout-aware posterior inference. Building and validating a 10-GHz FR3 RFSoC/PiRadio localization testbed that combines fixed and linear-track/D48 measurements with TurtleBot4 and Jackal UGV mobility, Vivaldi antennas, 2D LiDAR/RGB sensing, and Vicon ground truth under controlled LOS/NLOS transitions. This line has produced three 2026 conference papers—one at ***IEEE GLOBECOM*** and two at ***Asilomar***—with related journal manuscripts under review at ***IEEE TWC*** and ***IEEE TSP***.
- **Digital-twin priors for wireless-aware navigation**: Developed ray-tracing wireless digital twins and *PIRL* (physics-informed reinforcement learning) methods that use simulated propagation as a transferable prior for zero-shot indoor navigation, wireless SLAM, and symbolic-prior robot policies, with publications in ***ICRA***, ***IEEE OJ-COMS***, and ***RLC*** **(*Spotlight*)**; extending this line to physical localization/navigation experiments with TurtleBot4 and Jackal UGV platforms, FR3 channel-sounding hardware, and controlled navigation scenarios.

- **UE-centric twins for closed-loop multi-band adaptation**: Developing *MCMB-HDT* (Multi-Cell Multi-Band Handset Digital Twin), a UE-centric framework that couples real urban geometry, base-station topology, FR1/FR3 ray tracing, embodied antenna radiation, pedestrian motion, handset pose, and measurement-limited feedback for learning UE-side closed-loop array/band activation. Developed **Transformer**-based rate prediction from sparse asynchronous channel-quality/rate histories and, in digital-twin simulation, PPO-based retain-or-explore array activation that preserves best-link rate while penalizing extra-array measurements; the rate-prediction paper appears at ***IEEE ICC Workshops 2026***, and MCMB-HDT is under review at ***IEEE JSAC***.
- **Object-centric memory for multimodal spatial reasoning**: Developing adaptive sparse RGB/depth/pose sensing and queryable semantic 3D memory for wearable and embodied spatial intelligence. Building an object-centric dataset that links posed views, object records, visibility/occlusion, embeddings, and object–view/object–object relations for retrieval, localization, cross-view matching, spatial search, and agentic grounding, with selective edge/cloud updates.

NYU C2SMARTER Center — *Brooklyn, NY, USA*
Research Assistant — *Sept. 2023 – Mar. 2026*

- **Traffic mobility control**: Predictive correlated online learning and distributed pan-tilt-camera control for real-time traffic monitoring, estimation, and adaptive coverage; published in ***TR-C***.
- **Digital-twin closed loop**: Integration of loop detectors, video analytics, and control feedback for driver-risk-aware mobility/safety prediction and incident response; validated on a Brooklyn urban testbed, with work published in ***IEEE T-ITS***.
- **Distributed online control**: Distributed online convex optimization for networked sensing/control systems with nonseparable costs and time-varying constraints; published in ***IEEE L-CSS***.

NYU Center for Cybersecurity — *Brooklyn, NY, USA*
Research Assistant — *Sept. 2021 – Jan. 2024*

- **Safe adaptive control**: Game-theoretic, neuro-symbolic, and meta-RL methods for nonstationary driving, level-k pedestrian-aware autonomy, sampling-attack robustness, and distributed adaptive penetration testing; publications in ***ICRA*** and ***MILCOM***.
- **Route-recommendation control under strategic behavior**: Incentive-compatible route recommendation and proactive risk assessment for misinformation/demand attacks in navigation systems, combining Wardrop-equilibrium reasoning, Stackelberg-game modeling, and robust control; related work published in ***IEEE TIFS***.

PUBLICATIONS

Legend: **bold** = Haozhe Lei, * co-first author, † sole corresponding author.

Journal Articles

[J1] **Haozhe Lei**†, Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. “Site-Agnostic Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins”. *IEEE Transactions on Wireless Communications* (2026). Under review.

[J2] **Haozhe Lei**†, Hao Guo, and Sundeep Rangan. “Learning a Measurement-to-Posterior Map for Wireless Localization”. *IEEE Transactions on Signal Processing* (2026). Under review.

[J3] Ruibin Chen*, **Haozhe Lei***,†, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. “MCMB-HDT: A Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation”. *IEEE Journal on Selected Areas in Communications* (2026). Under review.

[J4] Zhaoye Pan, **Haozhe Lei**, Fan Zuo, Zilin Bian, and Tao Li. “Distributed Online Convex Optimization with Nonseparable Costs and Constraints”. *IEEE Control Systems Letters* 10 (2026), pp. 391–396. [PDF].

[J5] Tao Li*, Zilin Bian*, **Haozhe Lei***, Fan Zuo*, Ya-Ting Yang*, Quanyan Zhu, Zhenning Li, Zhibin Chen, and Kaan Ozbay. “Digital Twin-Based Driver Risk-Aware Predictive Mobility Analytics for Real-Time

Situational Awareness Through Cooperative Sensing". *IEEE Transactions on Intelligent Transportation Systems* (2025). [Article].

[J6] Tao Li, **Haozhe Lei**, Hao Guo, Mingsheng Yin, Yaqi Hu, Quanyan Zhu, and Sundeep Rangan. "Digital Twin-Enhanced Wireless Indoor Navigation: Achieving Efficient Environment Sensing with Zero-Shot Reinforcement Learning". *IEEE Open Journal of the Communications Society* (2025). [Article].

[J7] Ya-Ting Yang, **Haozhe Lei**, and Quanyan Zhu. "PRADA: Proactive Risk Assessment and Mitigation of Misinformed Demand Attacks on Navigational Route Recommendations". *IEEE Transactions on Information Forensics and Security* (2025). [PDF].

[J8] Zilin Bian, Tao Li, **Haozhe Lei**, Fan Zuo, Ya-Ting Yang, Quanyan Zhu, Zhenning Li, and Kaan Ozbay. "Multi-level Traffic-Responsive Tilt Camera Surveillance through Predictive Correlated Online Learning". *Transportation Research Part C: Emerging Technologies* (2024). [Article].

Conference Papers

[C1] **Haozhe Lei**†, Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. "LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins". *IEEE GLOBECOM*. 2026. [PDF].

[C2] **Haozhe Lei**†, Hao Guo, Tommy Svensson, and Sundeep Rangan. "Beyond Point Estimates: Likelihood-Based Full-Posterior Wireless Localization". *Asilomar*. 2026. [PDF].

[C3] Mingsheng Yin*, Tao Li*, **Haozhe Lei***, Yaqi Hu, Sundeep Rangan, and Quanyan Zhu. "Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning". *IEEE ICRA*. 2024. [PDF].

[C4] Tao Li*, **Haozhe Lei***, and Quanyan Zhu. "Self-Adaptive Driving in Nonstationary Environments through Conjectural Online Lookahead Adaptation". *IEEE ICRA*. 2023. [PDF].

[C5] Ruibin Chen*, **Haozhe Lei***,†, Hao Guo, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. "Transformer-Based Rate Prediction for Multi-Band Cellular Handsets". *IEEE ICC Workshops*. 2026. [PDF].

[C6] Roberto Bomfin, **Haozhe Lei**, Sundeep Rangan, and Marwa Chafii. "Likelihood-Based Wireless Localization with Last-Bounce Spatial Features". *Asilomar*. 2026.

[C7] Tao Li, **Haozhe Lei**, Mingsheng Yin, and Yaqi Hu. "Reinforcement Learning with Physics-Informed Symbolic Program Priors for Zero-Shot Wireless Indoor Navigation". *RLC*. Selected as a spotlight paper. 2025. [PDF].

[C8] **Haozhe Lei**, Yunfei Ge, and Quanyan Zhu. "ADAPT: A Game-Theoretic and Neuro-Symbolic Framework for Automated Distributed Adaptive Penetration Testing". *IEEE MILCOM*. 2024. [PDF].

TEACHING & MENTORING

New York University Spring 2024 – Present

Graduate Teaching

- ECE-GY5213 Introduction to System Engineering, Teaching Assistant Spring 2024

Graduate Mentoring

- **Ruibin Chen**, Ph.D. student, New York University *Feb. 2025 – Present*
  Project: Multi-Band Handset Digital Twins and Closed-Loop Array/Band Activation
  (IEEE ICC Workshops 2026; IEEE JSAC under review)
- **Yuheng Liu & Ashesh Kaji**, M.S. students, New York University *Jan. 2026 / Jul. 2026 – Present*
  Project: Multimodal Spatial Reasoning with Object-Centric Scene-Graph Memory and Agentic Inference
- **Yuhan Jiang**, M.S. student, New York University *Jul. 2026 – Present*
  Project: Robotic RF Measurement and FR3 Localization Testbed

INDUSTRY EXPERIENCE & RESEARCH COLLABORATION

Astrabeam LLC — *New York, NY, USA*
Research Collaborator — *Jun. 2025 – Aug. 2025*

- Collaborated on Doppler-based mmWave radar processing in ROS2 for odometry and object detection.

JD.com, Inc. — *Beijing, China*
Algorithm Engineer Intern — *Jan. 2021 – Feb. 2021*

- CenterNet (PIoU) rotational detection; on-site deployment to manufacturing lab.

AWARDS

- Ernst Weber Fellowship (NYU Tandon School of Engineering), 2023
- Tandon Research Excellence Exhibit [Link], Project: *AI-Driven Interactive Safe Autonomous Driving*, 2022
- School of Engineering Fellowship (NYU Tandon School of Engineering), 2022
- China National University Student Innovation & Entrepreneurship Training Program, 2017-2019

TECHNICAL EXPERTISE

- **Programming & Tools:** Python, Linux/Shell scripting, MATLAB, R, LaTeX, Kubernetes (HSRN cluster)
- **Simulation & Frameworks:** ROS2, Sionna, Wireless InSite, SUMO & CARLA (Transportation)
- **Robotics & Wireless Hardware:** TurtleBot4 mobile robot, Jackal mobile unmanned ground vehicle (UGV), RFSoC/PiRadio, D48 pan-tilt unit, linear track, TI mmWave radar (AWR1642, AWR6843), DCA1000

REVIEW SERVICES

Journals

- IEEE Transactions on Signal Processing (TSP)
- IEEE Transactions on Communications (TCOM)
- IEEE Wireless Communications Letters (WCL)
- IEEE Vehicular Technology Magazine (VTM)
- IEEE Control Systems Letters (L-CSS)

Conferences

- IEEE International Conference on Communications (ICC 2026)
- IEEE International Conference on Computer Communications (INFOCOM 2025)
- IEEE Global Communications Conference (GLOBECOM 2025)
- IEEE Conference on Decision and Control (CDC 2025)
- IEEE International Conference on Robotics and Automation (ICRA 2024)
- American Control Conference (ACC 2024)

REFERENCES

Prof. Sundeep Rangan, Director, NYU WIRELESS; Professor, NYU Tandon — srangan@nyu.edu
Prof. Ludovic Righetti, Professor, NYU Tandon; International Chair, ANITI — ludovic.righetti@nyu.edu
Prof. JohnRoss Rizzo, MD, Associate Professor, NYU Grossman School of Medicine — JohnRoss.Rizzo@nyulangone.org
Prof. Marco Mezzavilla, Associate Professor, Politecnico di Milano — marco.mezzavilla@polimi.it
Prof. Tao Li, Assistant Professor, City University of Hong Kong — li.tao@cityu.edu.hk